Notes: Acharya, Almeida, and Campello (JF, 2013)

Aggregate Risk and the Choice between Cash and Lines of Credit

Contents

1 Big picture

- **Question.** How should firms manage liquidity needs when the economy-wide risk of needing liquidity rises? In particular, when do firms rely on bank lines of credit and when do they instead self-insure by holding cash?
- **Core idea.** Bank credit lines are useful because banks can pool firms' idiosyncratic liquidity shocks. But banks cannot pool aggregate liquidity shocks: if many firms draw at once, credit-line commitments become risky for banks.
- **Main theory prediction.** Firms with more aggregate risk exposure should rely less on bank lines of credit and more on cash. Cash is costly because it earns a liquidity premium, but it is safer for firms whose liquidity needs are likely to arrive when many other firms also need liquidity.
- **Empirical object.** The paper studies the relative use of lines of credit versus cash, measured by *LC-to-Cash*: credit-line availability divided by credit-line availability plus cash.
- **Main cross-sectional result.** Firms with higher systematic risk, measured by asset betas, bank betas, tail betas, and cash-flow-based betas, have lower LC-to-Cash ratios.
- **Main pricing result.** Firms with higher aggregate risk exposure pay more for credit lines, especially through higher undrawn fees and higher all-in drawn spreads.
- **Main time-series result.** When aggregate risk is high, measured by VIX and Bank VIX, credit line initiations decline, credit line terms tighten, and firms accumulate more cash.
- **Mechanism result.** Bank risk rises with exposure to undrawn corporate credit lines only in high-VIX periods. This supports the model's claim that aggregate risk makes line-of-credit commitments risky for banks.

Economic takeaway: A credit line is not just "cash from the bank." It is insurance written by a financial intermediary. When the insured event is mostly idiosyncratic, banks can provide the insurance cheaply. When the insured event is aggregate, cash becomes more attractive because the bank's own capacity to honor commitments becomes fragile.

2 Incremental contribution of the paper

- **From liquidity level to liquidity composition.** Much of the earlier cash literature asks why firms hold liquidity. This paper asks how firms split liquidity between cash and credit lines.
- **A theory of cash versus committed credit.** The paper develops a model in which credit lines are bank-provided liquidity insurance and cash is self-insurance. The trade-off depends on aggregate risk exposure.
- **A banking-sector constraint channel.** The model gives banks an explicit role: credit lines are attractive when banks can pool risks, but aggregate liquidity shocks make bank commitments costly.
- **A new empirical prediction: beta matters for liquidity policy.** The paper predicts and finds that firms' systematic risk exposure affects the cash-credit-line mix, even after controlling for profitability, size, Q, net worth, tangibility, and volatility.

- **A pricing contribution.** The paper does not only show that high-beta firms use fewer credit lines; it also shows that their credit lines are more expensive.
- **A time-series contribution.** It links aggregate volatility to corporate liquidity management: when VIX or Bank VIX rises, firms initiate fewer credit lines and hold more cash.
- **A mechanism contribution.** The bank-level evidence shows that banks exposed to undrawn corporate credit lines become riskier in high aggregate-risk states, matching the model's liquidity-constraint mechanism.
- **A distinction from covenant-risk explanations.** Sufi (2009) emphasizes covenant violations and credit-line revocations. This paper shows that aggregate risk works through a different channel: the banking sector's ability to provide liquidity insurance.

3 Data and measurement

3.1 Credit line data

- **LPC-DealScan sample.** The large sample covers credit line initiations from 1987 to 2008. It is useful for broad cross-sectional and time-series tests but is tilted toward syndicated loans and larger firms.
- **Sufi random sample.** The smaller sample covers roughly 300 Compustat firms from 1996 to 2003 and includes detailed information on total and unused lines of credit. It is useful because it directly observes unused line availability.
- **Liquidity composition.** In the LPC sample, the main outcome is

$$LC\text{-}to\text{-}Cash_{it} = \frac{\text{Total LC}_{it}}{\text{Total LC}_{it} + \text{Cash}_{it}}.$$

  In Sufi's sample, the paper uses both total and unused credit lines in the numerator.

Interpretation: The LC-to-Cash ratio isolates the form of liquidity insurance. A firm may need more total liquidity because of investment opportunities or risk, but the ratio asks whether the marginal liquidity dollar is provided by the bank or held internally as cash.

3.2 Systematic risk measures

- **Beta KMV.** The benchmark measure is an unlevered asset beta constructed from equity betas using a Merton-KMV-style method.
- **Beta Asset.** An alternative asset beta from Choi (2009), based directly on asset returns.
- **Beta Cash.** An industry-level beta adjusted for cash holdings, designed to reduce the concern that high cash mechanically lowers asset beta.
- **Beta Bank.** A firm's beta with respect to a bank-stock index. This captures the firm's exposure to banking-sector downturns.
- **Beta Tail.** A downside beta based on firm performance when the stock market experiences its worst returns.

- **Beta Gap and Beta Cash Flow.** Cash-flow-based measures designed to capture whether a firm's financing needs or cash flows co-move with aggregate conditions.
- **SysVar KMV.** The systematic component of asset-return variance:

$$SysVar\,KMV_{jt} = (Beta\,KMV_{jt})^2 \times Var\,KMV_t.$$

Interpretation: The model's primitive is the correlation of liquidity needs across firms. The empirical measures approximate that primitive using market betas, bank-sector betas, downside betas, and cash-flow covariation.

3.3 Time-series and bank-level variables

- **VIX.** The paper uses VIX as a measure of aggregate risk and the market price/appetite for bearing that risk.
- **Bank VIX.** Because long historical implied-volatility data for banks are unavailable, the paper constructs expected banking-sector volatility using a GARCH model on bank stock returns.
- **Aggregate LC initiations.** The paper sums annual credit-line initiations and scales by aggregate assets.
- **Aggregate cash changes.** The paper measures aggregate changes in corporate cash holdings scaled by aggregate assets.
- **Bank Call Reports.** Bank-level tests use data on unused credit-line commitments and bank stock-return volatility.

Interpretation: The time-series tests ask whether aggregate volatility shifts the economy away from bank-provided liquidity and toward internal liquidity. The bank-level tests ask whether this is because undrawn credit commitments become riskier for banks.

4 Model and empirical specifications

4.1 Model structure

The model builds on Holmstrom and Tirole-style liquidity insurance. Firms invest at date 0, may face a liquidity need at date 1, and produce at date 2 only if the liquidity need is met.

- A fraction θ of firms have perfectly correlated liquidity shocks. These are systematic firms.
- The remaining $1-\theta$ firms have independent liquidity shocks. These are nonsystematic firms.
- The project has a moral-hazard friction, so not all future cash flow can be pledged to outsiders.
- Pledgeable income is

$$\rho_0 = p_G\left(R - \frac{B}{\Delta p}\right),$$

while the total expected payoff is

$$\rho_1 = p_G R.$$

- The key assumption is $\rho_0 < \rho < \rho_1$: the continuation investment is socially valuable, but pledgeable income alone cannot finance the liquidity shock.

Interpretation: The firm needs precommitted liquidity because waiting until the liquidity shock arrives is too late. Spot borrowing is insufficient when pledgeable income is below the liquidity need.

4.2 Credit lines versus cash

A line of credit is modeled as a bank commitment. The firm pays in good states and can draw in bad states. The bank can pool idiosyncratic needs but struggles when many firms draw at once.

The bank can provide first-best liquidity through lines of credit only if systematic risk is sufficiently low:

$$\theta < \theta^{max}.$$

When θ is large, systematic firms cannot fully rely on credit lines. They must hold cash or under-invest after liquidity shocks.

Economic intuition: Credit lines transform firm-level idiosyncratic liquidity risk into diversified bank risk. But if firm liquidity shocks are correlated, the bank is exposed to a large aggregate drawdown state.

4.3 Cross-sectional liquidity regressions

The benchmark empirical specification follows Sufi (2009) and adds systematic risk:

$$\begin{aligned} LC\text{-}to\text{-}Cash_{it} = \alpha &+ \beta_1 Beta\,KMV_{it} + \beta_2 \ln(Age)_{it} + \beta_3 Profitability_{i,t-1} \\ &+ \beta_4 Size_{i,t-1} + \beta_5 Q_{i,t-1} + \beta_6 Networth_{i,t-1} \\ &+ \beta_7 IndSalesVol_{jt} + \beta_8 ProfitVol_{it} + Year_t + \epsilon_{it}. \end{aligned}$$

The model predicts $\beta_1 < 0$: higher systematic risk should reduce reliance on credit lines relative to cash.

Interpretation: The regression asks whether two firms with similar size, profitability, Q, net worth, tangibility, age, and volatility choose different liquidity instruments because one has higher aggregate risk exposure.

4.4 Credit-line pricing regressions

To test whether aggregate-risky firms face worse line terms, the paper estimates regressions of the form

$$Cost_{it} = \mu_0 + \mu_1 Beta_{it} + \mu_2 \frac{LC_{it}}{Assets_{it}} + \mu_3 LIBOR_{it} + \mu_4 X_{it} + Year_t + \epsilon_{it}.$$

The dependent variables are all-in drawn spreads and undrawn fees. The model predicts $\mu_1 > 0$.

Interpretation: If banks charge more to high-beta firms, this directly supports the mechanism: high aggregate risk exposure makes liquidity insurance more expensive.

4.5 Time-series liquidity regressions

The paper estimates a seemingly unrelated regression system:

$$LCInitiation_t = \varsigma_0 + \varsigma_1 VIX_{t-1} + \varsigma_2 Trend_t + \varsigma_3 Controls_{t-1} + \varepsilon_t,$$

$$\Delta Cash_t = \gamma_0 + \gamma_1 VIX_{t-1} + \gamma_2 Trend_t + \gamma_3 Controls_{t-1} + u_t.$$

The model predicts $\varsigma_1 < 0$ and $\gamma_1 > 0$. The same logic applies to Bank VIX.

Interpretation: When aggregate risk rises, banks reduce the supply of new credit-line insurance and firms shift toward cash.

4.6 Mechanism tests

The bank-level mechanism regression links bank risk to credit-line commitments:

$$BankVol_{it} = \alpha + \beta Commitments_{i,t-1} + Controls + \epsilon_{it}.$$

The model predicts that β should be positive especially in high-VIX periods.

The paper also tests whether covenant violations or credit-line revocations explain the results. If the main mechanism were covenant risk, high-VIX periods or high-beta firms should have more violations or more revocations conditional on violations. The evidence does not support this alternative explanation.

5 Identification and empirical design

5.1 What the design can identify

- The cross-sectional tests identify whether systematic-risk exposure is associated with the liquidity composition of otherwise similar firms.
- The pricing tests identify whether high-systematic-risk firms face higher credit-line costs, conditional on firm and deal characteristics.
- The time-series tests identify whether aggregate volatility predicts shifts away from new credit lines and toward cash.
- The bank-level tests identify whether bank exposure to unused credit-line commitments is riskier when aggregate risk is high.

Interpretation: The evidence is not a randomized experiment. Its strength comes from a coherent set of mutually reinforcing predictions across firm choices, credit-line prices, aggregate time series, and bank risk.

5.2 Main identification concerns

- **Leverage mechanical effect.** Equity betas mechanically rise with leverage. The paper therefore uses unlevered asset betas and alternative asset-return betas.

- **Cash mechanical effect.** High cash holdings can mechanically lower asset beta. The paper uses cash-adjusted industry betas to address this.
- **Measurement error in beta.** Betas are noisy. The paper instruments beta proxies with their own lags and reports first-stage and overidentification diagnostics.
- **Demand for debt versus supply of credit lines.** The time-series evidence might reflect weak loan demand rather than bank constraints. The paper checks credit-line spreads and maturities and finds terms tighten when VIX rises.
- **Covenant violation channel.** Firms may avoid credit lines because they fear revocation after covenant violations. The paper shows that covenant violations are driven by profitability shocks, not aggregate risk exposure.

Interpretation: The paper's identification is strongest because the same mechanism predicts signs in many places: beta reduces LC-to-Cash, beta raises line costs, VIX reduces line initiations, VIX tightens line terms, and credit-line commitments raise bank risk only in high-VIX periods.

6 Tables and figures

6.1 Figure 1: Timeline of the Model; Figure 2: Equilibrium with Cash Holdings for Systematic Firms When Systematic Risk Is High

Read:

- Figure 1 lays out the model timing. Firms invest at date 0, then at date 1 may face either a liquidity need $\rho > 0$ or no liquidity need. Systematic firms face correlated shocks; other firms face idiosyncratic shocks.
- Figure 2 shows the equilibrium when aggregate risk is high. If the supply of liquid assets is sufficient, firms can sustain efficient continuation with partial cash reserves. If liquid-asset supply is low, a liquidity premium emerges and systematic firms partially liquidate after liquidity shocks.

Economic message: Cash is valuable because it is state-contingent self-insurance. Credit lines economize on cash only when the bank can pool risks across firms.

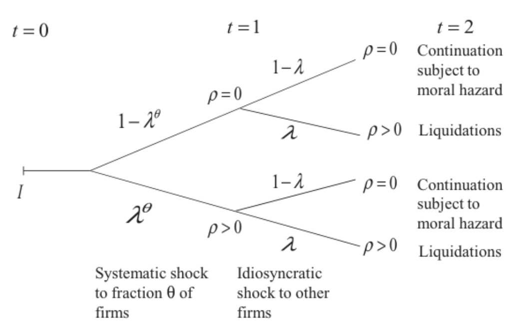

Figure 1. Timeline of the model.

(a) Figure 1: Timeline of the model

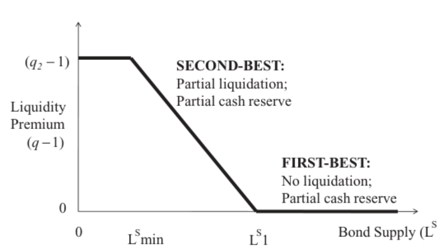

Figure 2. Equilibrium with cash holdings for systematic firms when systematic risk is high ($\theta \geq \theta^{\max}$).

(b) Figure 2: Equilibrium with cash holdings for systematic firms when systematic risk is high

6.2 Table I: Summary Statistics; Table II: Correlations

Read:

- Table I reports summary statistics for the LPC-DealScan and Sufi samples. The LPC sample is much larger and tilted toward larger firms; Sufi's sample is smaller but observes total and unused credit lines directly.
- Table II shows that the different market-based beta measures are positively correlated, while cash-flow-based betas are more weakly correlated with return-based betas.

Economic message: The paper has enough variation in liquidity composition and aggregate-risk exposure to test whether firms substitute between cash and credit lines.

Table I

Summary Statistics

This table reports summary statistics for empirical proxies related to firm characteristics: *LC-to-Cash*, *Assets*, *Tangibility*, *Q*, *NetWorth*, *Profitability*, *IndSaleVol*, *ProfitVol*, *Assets*, *Age*, *Unused LC-to-Cash*, *Total LC-to-Cash*, *Beta KMV*, *Beta Asset*, *Var KMV*, *Var Asset*, *Beta Cash*, *Beta Bank*, *Beta Tail*, *Beta Gap*, *Beta Cash Flow*, *Beta Equity*, *SysVar KMV*, *All-In Drawn Spread*, and *Undrawn Fee*. All variables are defined in Appendix F.

Variables	Mean	StDev	Median	Firm-years
	Panel A: LPC Credit Line Data			
LC-to-Cash	0.325	0.404	0.000	44,598
Tangibility	0.350	0.232	0.297	43,250
Assets	2,594	17,246	270	43,309
Q	1.961	1.314	1.475	43,288
Networth	0.381	0.248	0.404	43,288
Profitability	0.137	0.120	0.141	43,309
IndSalesVol	0.043	0.031	0.034	44,823
ProfitVol	0.063	0.053	0.044	44,821
Age	18.855	14.339	14.000	44,825
Beta KMV	0.986	1.032	0.856	44,402
Beta Cash	0.970	0.574	0.920	44,714
Beta Bank	0.445	0.703	0.390	44,440
Beta Tail	0.742	0.567	0.697	44,367
Beta Gap	0.906	1.420	0.681	35,532
Beta Cash Flow	0.926	1.397	0.697	35,532
Beta Equity	1.110	1.319	1.037	44,402
Beta Asset	0.919	0.926	0.756	14,646
Var KMV	0.017	0.019	0.009	44,825
Var Asset	0.012	0.017	0.005	14,646
SysVar KMV	0.013	0.018	0.006	44,402
All-In Drawn Spread	1.771	1.124	1.750	11,408
Undrawn Fee	0.315	0.167	0.300	9,865
	Panel B: Sufi Data			
Unused LC-to-Cash	0.450	0.373	0.455	1,906
Total LC-to-Cash	0.512	0.388	0.569	1,908
Unused LC-to-Assets	0.122	0.225	0.078	1,908
Covenant Violation	0.080	0.271	0.000	1,908
Tangibility	0.332	0.230	0.275	1,908
Assets	1,441	7,682	116	1,908
Q	2.787	3.185	1.524	1,905
Networth	0.426	0.300	0.453	1,905
Profitability	0.015	0.413	0.126	1,908
IndSalesVol	0.043	0.026	0.036	1,908
ProfitVol	0.089	0.078	0.061	1,908
Age	16.037	13.399	10.000	1,908
Beta KMV	1.002	1.068	0.804	1,559
Beta Cash	0.974	0.639	0.915	1,881
Beta Bank	0.479	0.756	0.400	1,561
Beta Tail	0.631	0.494	0.584	1,003
Beta Gap	1.000	1.755	0.892	1,677
Beta Cash Flow	0.904	1.549	0.733	1,677
Beta Equity	1.086	1.280	0.968	1,596
Beta Asset	0.957	0.995	0.705	643
Var KMV	0.026	0.026	0.015	1,568
Var Asset	0.023	0.025	0.011	643
SysVar KMV	0.019	0.023	0.008	1,559

(a) Table I: Summary Statistics

Table II

Correlations

This table shows the correlations for the different proxies for asset beta, idiosyncratic risk and systematic risk. See Appendix F for a description of the variables.

	Beta Asset	*Beta Cash*	*Beta Bank*	*Beta Tail*	*Beta Equity*	*Beta Gap*	*Beta Cash Flow*	*Beta KMV*	*Var KMV*	*SysVar KMV*
Beta Cash	0.4627									
Beta Bank	0.5591	0.2795								
Beta Tail	0.4685	0.3302	0.2215							
Beta Equity	0.817	0.371	0.6213	0.3124						
Beta Gap	0.1809	0.232	0.0955	0.0863	0.0849					
Beta Cash Flow	0.1684	0.196	0.0982	0.0828	0.0747	0.6114				
Beta KMV	0.8889	0.4502	0.6416	0.3865	0.9444	0.1131	0.0925			
Var KMV	0.4001	0.2165	0.2585	0.2	0.2804	0.1144	0.096	0.3803		
SysVar KMV	0.7086	0.3469	0.463	0.2859	0.596	0.1285	0.0977	0.7374	0.6337	
Var Assets	0.4414	0.238	0.2633	0.2584	0.2978	0.1511	0.1491	0.3872	0.9181	0.6149

(b) Table II: Correlations

6.3 Table III: The Choice between Cash and Credit Lines: LPC-Deal Scan Sample; Table IV: The Choice between Cash and Credit Lines: Sufi's (2009) Sample

Read:

- Table III uses the large LPC-DealScan sample. Beta KMV, SysVar KMV, and total asset variance are negatively related to LC-to-Cash, with the systematic component especially important.

- Table IV uses Sufi's sample and shows that the same beta relation holds for both total and unused credit-line measures.
- The controls replicate core findings from Sufi: more profitable and larger firms tend to rely more on credit lines, while high-Q and low-net-worth firms tend to rely less on them.

Economic message: Firms with greater aggregate risk exposure use less bank liquidity insurance and more cash, exactly as the model predicts.

Table III

The Choice between Cash and Credit Lines: LPC-Deal Scan Sample

This table reports regressions of a measure of line of credit use in corporate liquidity policy on proxies for asset beta, asset variance, and controls. The dependent variable is *LC-to-Cash*, defined in Appendix F. *Beta KMV* is the firm's asset (unlevered) beta, calculated from equity (levered) betas and a Merton-KMV formula. *Var KMV* is the corresponding value for total asset variance. *SysVar KMV* is a measure of firm-level systematic variance of asset returns. All proxies for Beta and variances are instrumented with their first two lags. All other variables are described in Appendix F. Robust t-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variable: *LC-to-Cash*				
	(1)	(2)	(3)	(4)	(5)
Beta KMV			−0.083***		−0.059*
			(−4.947)		(−1.778)
Var KMV		−3.920***			−1.681
		(−7.010)			(−1.209)
SysVar KMV				−6.213***	
				(−8.231)	
Profitability	0.136***	0.031	0.101***	0.044	0.063
	(5.435)	(0.959)	(3.274)	(1.396)	(1.633)
Tangibility	0.012	0.005	0.004	0.002	0.004
	(0.606)	(0.215)	(0.173)	(0.072)	(0.168)
Size	0.044***	0.039***	0.051***	0.047***	0.047***
	(16.151)	(12.285)	(16.151)	(16.045)	(8.726)
Networth	−0.138***	−0.146***	−0.132***	−0.133***	−0.136***
	(−9.817)	(−9.253)	(−8.008)	(−8.259)	(−7.883)
Q	−0.055***	−0.052***	−0.050***	−0.046***	−0.049***
	(−23.840)	(−18.017)	(−14.211)	(−14.156)	(−14.941)
IndSalesVol	−0.197	−0.206	−0.219	−0.202	−0.208
	(−1.343)	(−1.271)	(−1.349)	(−1.241)	(−1.279)
ProfitVol	−0.250***	0.148*	0.033	0.221**	0.121
	(−3.751)	(1.673)	(0.380)	(2.492)	(1.316)
Age	−0.047***	−0.054***	−0.052***	−0.056***	−0.053***
	(−7.933)	(−7.165)	(−6.819)	(−7.394)	(−7.049)
Constant	0.379***	0.568***	0.465***	0.514***	0.511***
	(5.710)	(7.307)	(6.044)	(6.704)	(6.064)
Industry fixed effect	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes
First-stage F-stat p-value		0.000	0.000	0.000	0.000
Hansen J-stat p-value		0.001	0.385	0.001	0.013
Observations	43,009	35,374	35,372	35,372	35,372
R^2	0.173	0.167	0.168	0.169	0.169

(a) Table III: The Choice between Cash and Credit Lines: LPC-Deal Scan Sample

Table IV

The Choice between Cash and Credit Lines: Sufi's (2009) Sample

This table reports regressions of a measure of line of credit usage in corporate liquidity policy on proxies for asset beta, asset variance, and controls. The dependent variables are *Unused LC-to-Cash* and *Total LC-to-Cash*, defined in Appendix F. *Beta KMV* is the firm's asset (unlevered) beta, calculated from equity (levered) betas and a Merton-KMV formula. *Var KMV* is the corresponding value for total asset variance. *SysVar KMV* is a measure of firm-level systematic variance of asset returns. All proxies for Beta and variances are instrumented with their first two lags. All other variables are described in Appendix F. Robust t-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variable: *Total LC-to-Cash*				Dependent Variable: *Unused LC-to-Cash*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Beta KMV		−0.336***	−0.419***			−0.270***	−0.322**	
		(−5.489)	(−2.801)			(−4.893)	(−2.438)	
Var KMV			3.114				1.649	
			(0.654)				(0.387)	
SysVar KMV				−17.119***				−15.026***
				(−5.789)				(−5.327)
Profitability	0.078**	−0.013	0.003	−0.004	0.061*	−0.012	−0.004	−0.008
	(2.269)	(−0.226)	(0.052)	(−0.083)	(1.955)	(−0.238)	(−0.074)	(−0.176)
Tangibility	0.040	−0.089	−0.081	−0.110	0.025	−0.091	−0.088	−0.113
	(0.560)	(−1.098)	(−0.938)	(−1.318)	(0.371)	(−1.184)	(−1.092)	(−1.439)
Size	0.047***	0.071***	0.083***	0.055***	0.053***	0.074***	0.081***	0.062***
	(5.110)	(5.593)	(3.621)	(4.593)	(6.106)	(6.481)	(3.992)	(5.687)
Networth	−0.097**	−0.077	−0.072	−0.071	−0.054	−0.043	−0.040	−0.036
	(−2.293)	(−1.345)	(−1.141)	(−1.231)	(−1.396)	(−0.819)	(−0.708)	(−0.678)
Q	−0.036***	−0.019***	−0.016	−0.018**	−0.029***	−0.016**	−0.013	−0.013*
	(−8.495)	(−2.656)	(−1.516)	(−2.472)	(−7.263)	(−2.398)	(−1.479)	(−1.918)
IndSalesVol	1.094*	−0.156	−0.138	−0.345	1.042	−0.073	−0.075	−0.278
	(1.691)	(−0.215)	(−0.186)	(−0.484)	(1.549)	(−0.093)	(−0.095)	(−0.355)
ProfitVol	−0.596***	0.315	0.272	0.548*	−0.554***	0.198	0.192	0.461
	(−3.209)	(1.022)	(0.887)	(1.694)	(−3.162)	(0.711)	(0.716)	(1.512)
Age	−0.039*	−0.086***	−0.083***	−0.097***	−0.023	−0.061**	−0.061**	−0.074**
	(−1.846)	(−2.818)	(−2.731)	(−3.028)	(−1.125)	(−2.101)	(−2.102)	(−2.438)
Constant	0.748***	0.306**	0.250	0.294**	0.148	0.165	0.141	0.172
	(8.612)	(2.359)	(1.516)	(2.306)	(1.377)	(1.332)	(0.945)	(1.404)
Ind. fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat p-val.		0.000	0.000	0.000	0.000	0.000	0.000	0.000
Hansen J-stat p-value		0.283	0.788		0.059	0.174	0.296	0.033
Observations	1,905	1,321	1,321	1,321	1,903	1,319	1,319	1,319
R^2	0.401	0.437	0.444	0.445	0.3713	0.399	0.406	0.411

(b) Table IV: The Choice between Cash and Credit Lines: Sufi's (2009) Sample

6.4 Table V: The Choice between Cash and Credit Lines: Varying Betas

Read:

- Table V replaces the benchmark Beta KMV with a wide set of beta measures: Beta Asset, Beta Cash, Beta Bank, Beta Tail, Beta Equity, Beta Gap, Beta Cash Flow, and industry Beta KMV.
- The coefficient on beta is negative across specifications. The bank-beta result is especially important because it shows that exposure to banking-sector risk predicts greater reliance on cash.
- Tail beta also matters: firms that perform poorly in market downturns use fewer credit lines relative to cash.

Economic message: The result is not an artifact of one beta definition. Many measures of systematic liquidity risk point to the same cash-versus-credit-line trade-off.

Table V

The Choice between Cash and Credit Lines: Varying Betas

This table reports regressions of a measure of line of credit use in corporate liquidity policy on asset (unlevered) beta and controls. *Beta KMV (Industry)* is the value-weighted average (three-digit SIC) industry *Beta KMV*. All other variables are described in Appendix F. In columns (1) to (7), beta measures are instrumented with their first two lags. In column (8), we use an industry beta rather than the firm-level instrumented beta in the regression. Robust t-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variable: *LC-to-Cash*							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Beta Asset	−0.137***							
	(−6.006)							
Beta Cash		−0.127***						
		(−9.258)						
Beta Bank			−0.291***					
			(−4.952)					
Beta Tail				−0.137***				
				(−7.417)				
Beta Equity					−0.054***			
					(−3.453)			
Beta Gap						−0.010***		
						(−3.428)		
Beta Cash Flow							−0.013***	
							(−4.515)	
Beta KMV (Industry)								−0.029***
								(−4.919)

(Continued)

(a) Table V: The Choice between Cash and Credit Lines: Varying Betas, part 1

Table VI—*Continued*

	Dependent Variable: *LC-to-Cash*							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Profitability	0.042	0.116***	0.078**	0.128***	0.113***	0.117***	0.118***	0.124***
	(0.675)	(5.088)	(2.284)	(4.364)	(3.878)	(4.779)	(4.853)	(5.008)
Tangibility	−0.017	−0.004	−0.015	−0.001	0.006	0.025	0.026	0.048**
	(−0.400)	(−0.239)	(−0.629)	(−0.027)	(0.269)	(1.320)	(1.372)	(2.400)
Size	0.042***	0.050***	0.055***	0.059***	0.052***	0.049***	0.049***	0.042***
	(6.965)	(19.963)	(15.334)	(16.679)	(15.804)	(17.865)	(17.905)	(14.520)
Networth	−0.127***	−0.109***	−0.120***	−0.118***	−0.149***	−0.124***	−0.125***	−0.114***
	(−4.169)	(−8.612)	(−6.836)	(−7.169)	(−9.720)	(−9.080)	(−9.162)	(−8.204)
Q	−0.050***	−0.049***	−0.048***	−0.044***	−0.053***	−0.056***	−0.056***	−0.052***
	(−8.136)	(−23.028)	(−12.198)	(−12.330)	(−17.681)	(−25.415)	(−25.507)	(−22.093)
IndSalesVol	−0.341	−0.128	−0.156	−0.174	−0.190	−0.187	−0.172	0.132
	(−1.029)	(−1.066)	(−0.936)	(−1.063)	(−1.207)	(−1.356)	(−1.255)	(0.826)
ProfitVol	−0.315*	−0.013	0.120	0.065	−0.065	−0.254***	−0.254***	−0.198***
	(−1.747)	(−0.199)	(1.168)	(0.797)	(−0.780)	(−3.608)	(−3.592)	(−2.785)
Age	−0.029**	−0.048***	−0.054***	−0.053***	−0.053***	−0.042***	−0.043***	−0.046***
	(−2.064)	(−8.494)	(−6.916)	(−7.005)	(−7.535)	(−6.678)	(−6.710)	(−6.902)
Constant	0.448**	0.614***	0.468***	0.415***	0.477***	0.453***	0.454***	0.362***
	(2.529)	(21.435)	(5.800)	(5.613)	(6.472)	(16.661)	(16.738)	(13.516)
Industry fixed effect	Yes	No	Yes	Yes	Yes	No	Yes	No
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
Hansen J-stat p-value	0.158	0.005	0.586	0.000	0.156	0.8728	0.001	
Observations	9,536	46,865	35,499	35,343	38,397	37,485	37,485	31,811
	31,811							
R^2	0.211	0.162	0.166	0.170	0.166	0.155	0.155	0.166

(b) Table V: The Choice between Cash and Credit Lines: Varying Betas, part 2

6.5 Table VI: Sorting on Proxies for Financing Constraints and Beta

Read:

- Table VI sorts firms by size, payout, ratings, and beta. The negative relation between beta and LC-to-Cash is concentrated among financially constrained firms.
- The beta effect is also strongest among high-beta firms. This matches the model's nonlinear prediction: when systematic risk is low, bank liquidity constraints need not bind; when systematic risk is high, they bind and force a shift toward cash.
- Panel B using Beta Tail produces similar evidence, especially for small, low-payout, nonrated, and high-beta firms.

Economic message: The cash-credit-line trade-off is economically relevant when firms need pre-committed liquidity and cannot easily raise external finance on the spot market.

Table VI

Sorting on Proxies for Financing Constraints and Beta

This table reports regressions of a measure of line of credit use in corporate liquidity policy on different proxies for asset beta and controls. The dependent variable is *LC-to-Cash*, defined in Appendix F. *Beta KMV* is the firm's asset (unlevered) beta, calculated from equity (levered) betas and a Merton-KMV formula. *Var KMV* is the corresponding value for total asset variance. *Beta Tail* is a measure of beta that is based on the average stock return of a firm in the days in which the stock market had its worst 5% returns in the year. All beta and variance measures are instrumented with their first two lags. In column (1) we use a sample of small firms (those with *Assets* in the 30th percentile and lower). In column (2) we use a sample of large firms (those with *Assets* in the 70th percentile and higher). In column (3) we use a sample of firms with low payouts (those with payout in the 30th percentile and lower). In column (4) we use a sample of firms with high payouts (those with payout in the 70th percentile and higher). In column (5) we use a sample of firms that have neither a bond nor a commercial paper rating. In column (6) we use a sample of firms that have both bond and commercial paper ratings. In column (7) we use a sample of firms with beta greater than one. In column (8) we use a sample of firms with beta lower than one. Panel A reports results using *Beta KMV*, and Panel B reports results using *Beta Tail*. All other variables are described in Appendix F. Robust t-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

	(1) Small firms	(2) Large firms	(3) Low payout firms	(4) High payout firms	(5) Non-rated firms	(6) Rated firms	(7) High beta firms	(8) Low beta firms
	Panel A: Beta KMV							
Beta KMV	−0.236**	−0.002	−0.191***	0.027	−0.069	0.092	−0.348***	0.067
	(−2.077)	(−0.029)	(−3.483)	(0.492)	(−1.467)	(0.789)	(−4.188)	(0.390)
Var KMV	6.628	−6.751**	2.739	−4.972**	−0.634	−14.317*	5.013*	−2.759**
	(1.523)	(−2.387)	(1.260)	(−2.170)	(−0.329)	(−1.706)	(1.883)	(−2.559)
Profitability	0.136*	0.156	0.217***	−0.039	0.029	0.038	0.075	0.067
	(1.663)	(1.560)	(3.882)	(−0.640)	(0.670)	(0.155)	(1.488)	(1.138)
Tangibility	−0.011	−0.003	−0.011	0.036	0.013	0.007	0.001	0.007
	(−0.338)	(−0.080)	(−0.404)	(1.112)	(0.532)	(0.089)	(0.024)	(0.265)
Size	0.109***	0.004	0.072***	0.036***	0.056***	0.010	0.043***	0.042***
	(4.655)	(0.447)	(7.508)	(4.897)	(6.109)	(0.496)	(7.488)	(3.710)
Networth	−0.060**	−0.186***	−0.080***	−0.174***	−0.119***	−0.267***	−0.101***	−0.183***
	(−2.008)	(−4.821)	(−3.463)	(−6.371)	(−6.235)	(−3.478)	(−5.515)	(−6.988)
Q	−0.006	−0.066***	−0.026***	−0.053***	−0.044***	−0.054***	−0.042***	−0.054***
	(−0.458)	(−9.372)	(−4.273)	(−10.778)	(−10.392)	(−3.140)	(−10.105)	(−13.284)

(Continued)

(a) Table VI: Sorting on Proxies for Financing Constraints and Beta, part 1

Table VI—*Continued*

	(1) Small firms	(2) Large firms	(3) Low payout firms	(4) High payout firms	(5) Non-rated firms	(6) Rated firms	(7) High beta firms	(8) Low beta firms
	Panel A: Beta KMV							
IndSalesVol	0.188	−0.149	0.000	−0.581**	−0.090	0.104	−0.200	−0.190
	(0.668)	(−0.474)	(0.002)	(−2.454)	(−0.494)	(0.199)	(−1.020)	(−0.971)
ProfitVol	−0.201	0.365*	−0.047	0.192	0.164	0.154	0.075	0.148
	(−1.012)	(1.732)	(−0.394)	(1.266)	(1.611)	(0.249)	(0.666)	(1.162)
Age	−0.009	−0.039***	−0.037***	−0.051***	−0.054***	−0.048*	−0.049***	−0.060***
	(−0.521)	(−2.995)	(−3.741)	(−4.793)	(−6.044)	(−1.800)	(−5.357)	(−6.623)
Constant	−0.046	0.819***	0.293**	0.548***	0.437***	0.551***	0.851***	0.487***
	(−0.231)	(5.966)	(2.239)	(4.288)	(4.389)	(2.389)	(9.623)	(5.031)
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Hansen J-stat p-value	0.913	0.001	0.284	0.012	0.400	0.262	0.050	0.498
Observations	8,436	12,578	14,908	14,162	22,546	4,344	15,212	20,160
R^2	0.105	0.148	0.182	0.170	0.138	0.164	0.187	0.187

(Continued)

(b) Table VI: Sorting on Proxies for Financing Constraints and Beta, part 2

Table VI—*Continued*

	(1) Small firms	(2) Large firms	(3) Low payout firms	(4) High payout firms	(5) Non-rated firms	(6) Rated firms	(7) High beta firms	(8) Low beta firms
	Panel B: Beta Tail							
Beta Tail	−0.228***	−0.019	−0.178***	−0.091**	−0.151***	0.077	−0.574***	−0.162**
	(−4.518)	(−0.385)	(−4.452)	(−2.481)	(−4.556)	(0.736)	(−4.793)	(−2.525)
Var KMV	2.578*	−6.301***	−0.305	−1.787	0.199	−12.204*	1.145	−1.265
	(1.833)	(−3.194)	(−0.260)	(−1.319)	(0.193)	(−1.766)	(0.654)	(−1.260)
Profitability	0.133***	0.161	0.207***	0.013	0.077*	0.061	0.043	0.164***
	(2.621)	(1.589)	(4.448)	(0.228)	(1.943)	(0.248)	(0.919)	(3.255)
Tangibility	−0.010	−0.005	−0.012	0.030	0.005	0.008	−0.003	−0.010
	(−0.353)	(−0.122)	(−0.461)	(0.928)	(0.221)	(0.098)	(−0.101)	(−0.379)
Size	0.109***	0.006	0.070***	0.048***	0.069***	0.006	0.035***	0.064***
	(8.191)	(0.541)	(9.223)	(7.639)	(9.636)	(0.301)	(5.622)	(8.554)
Networth	−0.076***	−0.182***	−0.083***	−0.150***	−0.103***	−0.276***	−0.064***	−0.143***
	(−3.878)	(−4.730)	(−4.063)	(−5.829)	(−5.789)	(−3.430)	(−3.042)	(−6.633)
Q	−0.003	−0.066***	−0.025***	−0.048***	−0.036***	−0.061***	−0.038***	−0.051***
	(−0.520)	(−9.072)	(−4.638)	(−9.650)	(−8.922)	(−4.136)	(−8.973)	(−12.136)
IndSalesVol	0.155	−0.142	0.156	−0.560**	−0.073	0.052	0.056	−0.296
	(0.592)	(−0.450)	(0.724)	(−2.370)	(−0.406)	(0.100)	(0.247)	(−1.599)
ProfitVol	0.012	0.375*	0.079	0.161	0.186**	0.179	0.195	0.146
	(0.095)	(1.793)	(0.773)	(1.062)	(1.972)	(0.291)	(1.606)	(1.297)
Age	−0.023*	−0.038***	−0.043***	−0.050***	−0.057***	−0.050*	−0.047***	−0.058***
	(−1.679)	(−2.956)	(−4.465)	(−4.736)	(−6.458)	(−1.852)	(−4.391)	(−6.768)
Constant	0.047	0.810***	0.334***	0.474***	0.358***	0.600***	1.203***	0.399***
	(0.404)	(5.787)	(2.820)	(3.693)	(3.921)	(2.369)	(10.152)	(4.253)
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Hansen J-stat p-value	0.084	0.001	0.000	0.076	0.000	0.345	0.000	0.0223
Observations	8,418	12,573	14,892	14,151	22,528	4,344	10,679	24,664
R^2	0.108	0.149	0.184	0.171	0.140	0.164	0.191	0.160

(c) Table VI: Sorting on Proxies for Financing Constraints and Beta, part 3

6.6 Figure 3: Average LC-Cash Ratios for Different Quintiles of Beta KMV; Table VII: Aggregate Risk Exposure and the Cost of Credit Lines

Read:

- Figure 3 shows that LC-to-Cash is fairly flat across low and middle beta quintiles but falls sharply in the highest beta quintile.
- Table VII shows that aggregate-risk exposure raises credit-line costs. Beta KMV, SysVar KMV, and Beta Tail are associated with higher undrawn fees and generally higher drawn spreads.

Economic message: The quantity and price evidence line up. High-beta firms use fewer credit lines because banks charge more to insure them against liquidity shocks.

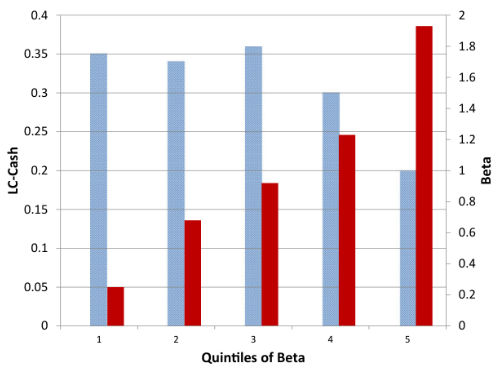

Figure 3. Average *LC-Cash* ratios for different quintiles of *Beta KMV*. This figure reports the average *LC-Cash* ratio for firms in different quintiles of *Beta KMV*. The sample is sorted into quintiles based on the average value of *Beta KMV* for each firm during the entire sample period. We then calculate the average value of the *LC-Cash* ratio in each of these quintiles of beta.

on systematic risk proxies and controls. We control for the size of credit line

(a) Figure 3: Average LC-Cash ratios for different quintiles of Beta KMV

Table VII

Aggregate Risk Exposure and the Cost of Credit Lines

This table reports regressions of line of credit spreads and fees on systematic risk exposure and controls. *LIBOR* is the level of the LIBOR (in percentage) in the quarter in which a deal was initiated, for each firm. *New LC* is the total size of deals initiated in a firm-year, scaled by assets. All other variables are described in Appendix F. All proxies for beta are instrumented with their first two lags. Robust z-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variables					
	All-In Drawn Spread			*Undrawn Fee*		
	(1)	(2)	(3)	(4)	(5)	(6)
Beta KMV	0.104			0.051***		
	(1.525)			(3.560)		
SysVar KMV		10.029***			3.124***	
		(2.740)			(4.047)	
Beta Tail			0.142**			0.047***
			(2.347)			(4.077)
LIBOR	−0.005	−0.005	−0.001	0.003	0.001	0.005
	(−0.242)	(−0.230)	(−0.051)	(0.631)	(0.188)	(1.291)
New LC	−0.236***	−0.237***	−0.240***	−0.018***	−0.020***	−0.020***
	(−3.355)	(−3.555)	(−3.503)	(−5.102)	(−6.620)	(−6.954)
Profitability	−1.852***	−1.732***	−1.897***	−0.095***	−0.073**	−0.120***
	(−11.885)	(−10.871)	(−12.435)	(−3.163)	(−2.359)	(−4.351)
Tangibility	0.131**	0.142**	0.140**	0.027**	0.030**	0.030**
	(2.192)	(2.390)	(2.367)	(2.205)	(2.451)	(2.555)
Size	−0.368***	−0.362***	−0.376***	−0.046***	−0.044***	−0.049***
	(−45.526)	(−48.126)	(−41.431)	(−27.482)	(−26.389)	(−26.738)
Networth	−1.212***	−1.217***	−1.237***	−0.193***	−0.188***	−0.196***
	(−20.204)	(−21.222)	(−20.412)	(−17.657)	(−18.006)	(−18.657)
Q	−0.150***	−0.164***	−0.154***	−0.036***	−0.038***	−0.035***
	(−10.034)	(−10.493)	(−11.052)	(−12.703)	(−12.967)	(−14.002)
IndSalesVol	0.238	0.219	0.131	−0.057	−0.046	−0.099
	(0.521)	(0.488)	(0.291)	(−0.629)	(−0.512)	(−1.149)
ProfitVol	2.266***	1.823***	2.275***	0.128*	0.047	0.173***
	(6.337)	(4.695)	(7.392)	(1.880)	(0.623)	(2.978)
Constant	4.790***	4.726***	4.828***	0.668***	0.659***	0.683***
	(21.626)	(22.075)	(22.695)	(18.972)	(19.000)	(21.034)
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Fst.-stage F-stat p-val	0.000	0.000	0.000	0.000	0.000	0.000
Hansen F-stat p-value	0.017	0.082	0.652	0.942	0.753	0.381
Observations	6,799	6,895	6,774	5,977	6,084	5,973
R^2	0.559	0.550	0.559	0.404	0.405	0.405

(b) Table VII: Aggregate Risk Exposure and the Cost of Credit Lines

6.7 Figure 4: Aggregate Risk and Time-Series Changes in Cash and Credit Line Initiations; Table VIII: Aggregate Risk and the Choice between Cash and Credit Lines: Time-Series Tests

Read:

- Figure 4 plots VIX, changes in cash, and credit line initiations. Credit line initiations tend to fall when VIX rises.
- Table VIII confirms this relation in time-series regressions. VIX and Bank VIX reduce credit-line initiations, while VIX raises changes in cash.
- The residual specifications show that the pattern survives after removing the effect of observable firm characteristics.

Economic message: When aggregate risk rises, firms shift away from new bank liquidity insurance and toward internal liquidity.

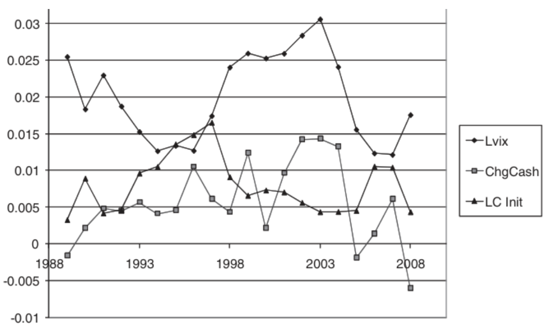

Figure 4. Aggregate risk and time-series changes in cash and credit line initiations. This figure reports changes in aggregate credit line initiations and changes in aggregate cash holdings. *LC Initiations* is defined as the sum of all credit line initiations in the LPC-Deal Scan sample in a given year, scaled by aggregate assets. *Change in Cash* is defined as the change in aggregate cash holdings in the LPC-Deal Scan sample scaled by aggregate assets. *VIX* is the implied volatility on S&P 500 index options, lagged one period (VIX is divided by 10 in this figure).

(a) Figure 4: Aggregate risk and time-series changes in cash and credit line initiations

Table VIII

Aggregate Risk and the Choice between Cash and Credit Lines: Time-Series Tests

This table reports regressions of aggregate credit line initiations and changes in aggregate cash holdings on macroeconomic variables. We estimate the SUR (seemingly unrelated regression model) in equation (A5) in the text. The dependent variable in columns (1) to (3) of Panel A is *LC Initiations*, which is defined as the sum of all credit line initiations in the LPC-Deal Scan sample in a given year, scaled by aggregate assets. The dependent variable in columns (1) to (3) in Panel B is *Change in Cash*, which is defined as the change in aggregate cash holdings in the LPC-Deal Scan sample scaled by aggregate assets; see equation (28) in the text. The dependent variable in columns (4) to (6) of Panel A (Panel B) is the average residual value of *LC Initiations* (*Change in Cash*) after controlling for the firm characteristics in equation (A3), excluding beta and year fixed effects. The independent variables are *VIX*, the implied volatility on S&P 500 index options, *BankVIX*, the expected volatility on an index of bank stock returns (computed using a GARCH model), *CP Spread*, the three-month commercial paper-Treasury spread, *Real GDP Growth*, and a time trend. All independent variables are lagged one period. Robust z-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

Panel A: Credit Line Tests

	Dependent Variables					
	LC Initiations			*Resid. LC Init.*		
	(1)	(2)	(3)	(4)	(5)	(6)
VIX_{t-1}	−0.040***		−0.022**	−0.054***		−0.022
	(−3.856)		(−2.259)	(−2.632)		(−1.064)
$Bank\ VIX_{t-1}$		−0.033***	−0.024***		−0.053***	−0.044***
		(−4.944)	(−3.463)		(−4.122)	(−3.005)
$CP\ Spread_{t-1}$	0.002	0.007**	0.006*	0.001	0.010	0.008
	(0.459)	(2.116)	(1.887)	(0.122)	(1.503)	(1.309)
$Real\ GDP\ Growth_{t-1}$	0.087*	−0.013	0.013	0.159	−0.002	0.024
	(1.756)	(−0.261)	(0.300)	(1.592)	(−0.022)	(0.252)
$Time\ Trend_{t-1}$	−0.066	0.282**	0.192*	0.429**	0.989***	0.899***
	(−0.631)	(2.430)	(1.726)	(2.051)	(4.412)	(3.847)
Constant	0.014***	0.020***	0.021***	0.000	0.013*	0.013**
	(4.512)	(5.733)	(6.580)	(0.023)	(1.860)	(1.989)
Observations	20	20	20	20	20	20
R^2	0.487	0.598	0.679	0.404	0.566	0.589

Panel B: Cash Tests

	Dependent Variables					
	Chg. Cash			*Resid. Chg. Cash*		
	(1)	(2)	(3)	(4)	(5)	(6)
VIX_{t-1}	0.040**		0.054***	0.092**		0.060
	(2.213)		(2.642)	(2.358)		(1.367)
$Bank\ VIX_{t-1}$		0.002	−0.019		0.066**	0.042
		(0.120)	(−1.299)		(2.328)	(1.321)
$CP\ Spread_{t-1}$	−0.003	−0.003	0.001	0.006	−0.005	−0.001
	(−0.448)	(−0.383)	(0.078)	(0.449)	(−0.342)	(−0.085)
$Real\ GDP\ Growth_{t-1}$	−0.084	−0.078	−0.143	0.057	0.258	0.186
	(−0.959)	(−0.728)	(−1.498)	(0.301)	(1.246)	(0.907)
$Time\ Trend_{t-1}$	0.058	0.039	0.265	0.223	−0.478	−0.227
	(0.316)	(0.150)	(1.114)	(0.565)	(−0.963)	(−0.447)
Constant	0.000	0.007	0.006	−0.024**	−0.035**	−0.036**
	(0.043)	(0.944)	(0.885)	(−2.086)	(−2.303)	(−2.509)
Observations	20	20	20	20	20	20
R^2	0.240	0.054	0.299	0.240	0.235	0.301

(b) Table VIII: Aggregate Risk and the Choice between Cash and Credit Lines: Time-Series Tests

6.8 Table IX: Aggregate Risk, Credit Line Contractual Terms, and Changes in Total Debt; Figure 5: Aggregate Risk and Time Series Changes in Credit Line Contractual Terms

Read:

- Table IX shows that VIX shortens average credit-line maturity and raises average spreads. Bank VIX also shortens maturity.
- The same table shows that aggregate risk does not simply reduce all debt issuance; changes in total debt are not negatively related to VIX in the same way.
- Figure 5 visualizes the tightening of credit-line terms when aggregate risk increases.

Economic message: The results look like a reduction in the supply of credit-line insurance, not just a reduction in firms' general demand for debt.

Table IX
Aggregate Risk, Credit Line Contractual Terms, and Changes in Total Debt

This table reports regressions of credit line contractual terms (maturity and spreads) and changes in aggregate debt on macroeconomic variables. In columns (1) and (2), and (3) and (4), we estimate the SUR (seemingly unrelated regression model) in equation (CI) in the text. The dependent variable in columns (1) and (3) is *Average Maturity*, which is defined as the average maturity (weighted by the size of the facility) in the LPC-Deal Scan sample for each year in the sample period. The dependent variable in columns (2) and (4) is *Average Spread*, which is defined as the average all-in-drawn spread (weighted by the size of the facility) in the LPC-Deal Scan sample for each year in the sample period. In columns (5) and (6) we estimate a SUR model such as that in equation (A5) in the text, replacing *LCInitiations* with *ChangeinDebt*, which is defined in equation (C2) in the text. The variable represents the aggregate change in total debt (short plus long term) in the LPC-Deal Scan sample for each year, scaled by aggregate assets. The independent variables are *VIX*, the implied volatility on S&P 500 index options, *BankVIX*, the expected volatility on an index of bank stock returns (computed using a GARCH model), *CPspread*, the three-month commercial paper-Treasury spread, *Real GDP Growth*, and a time trend. All independent variables are lagged one period. Robust z-statistics are presented in parentheses. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variables					
	Avg. Maturity (1)	Avg. Spread (2)	Avg. Maturity (3)	Avg. Spread (4)	Agg. Change in Total Debt (5)	Agg. Change in Total Debt (6)
$VIX_{t\text{-}1}$	−26.192***	1.756**			0.026	
	(−3.808)	(2.284)			(0.668)	
$Bank\ VIX_{t\text{-}1}$			−12.269**	0.733		−0.011
			(−2.063)	(1.214)		(−0.390)
$CP\ Spread_{t\text{-}1}$	−4.475*	0.500*	−2.483	0.382	0.025*	0.027*
	(−1.870)	(1.869)	(−0.817)	(1.238)	(1.809)	(1.832)
Real GDP $Growth_{t\text{-}1}$	55.280*	−4.497	17.584	−2.243	0.601***	0.567***
	(1.661)	(−1.209)	(0.403)	(−0.506)	(3.178)	(2.709)
$Time\ Trend_{t\text{-}1}$	−184.984***	2.416	−53.980	−5.414	−0.787**	−0.669
	(−2.650)	(0.310)	(−0.516)	(−0.510)	(−1.984)	(−1.333)
Constant	17.892***	0.567**	18.138***	0.590*	−0.002	0.008
	(8.898)	(2.523)	(5.732)	(1.836)	(−0.140)	(0.556)
Observations	20	20	20	20	20	20
R^2	0.582	0.328	0.405	0.211	0.516	0.509

average maturities and spreads are used as dependent variables:

(a) Table IX: Aggregate Risk, Credit Line Contractual Terms, and Changes in Total Debt

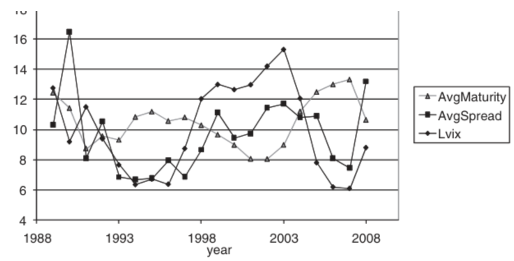

Figure 5. Aggregate risk and time series changes in credit line contractual terms. This table reports over-time changes in credit line contractual terms (maturity and spreads). *Average Maturity* is defined as the average maturity (weighted by the size of the facility) in the LPC-Deal Scan sample for each year in the sample period. *Average Spread* is defined as the average all-in-drawn spread (weighted by the size of the facility) in the LPC-Deal Scan sample for each year in the sample period. It is expressed in basis points and divided by 10. *VIX* is the implied volatility on S&P 500 index options, lagged one period. *VIX* is expressed in percentage points, and divided by two.

(b) Figure 5: Aggregate risk and time series changes in credit line contractual terms

6.9 Table X: Aggregate Risk and Banks' Liquidity Constraints; Table XI: Covenant Violations and Aggregate Risk

Read:

- Table X tests the bank-side mechanism. During high-VIX periods, banks with more unused corporate credit-line commitments have higher stock-return volatility. During low-VIX periods, this relation disappears.
- Table XI tests whether covenant violations rise with aggregate risk. Profitability shocks strongly predict covenant violations, but VIX and beta do not explain violations in a way that would account for the paper's main results.

Economic message: The bank evidence supports the model's liquidity-constraint channel. The covenant evidence weakens an alternative explanation based on revocation risk.

Table X

Aggregate Risk and Banks' Liquidity Constraints

This table reports regressions of bank stock return volatility on bank exposure to undrawn corporate credit lines, and other bank-level and macroeconomic variables. The variable definitions follow Gatev, Schuermann, and Strahan (2009). The dependent variable is the annualized monthly average of bank squared returns. *Commitments*$_{t-1}$ is the lagged ratio of undrawn credit lines for each bank, divided by the sum of undrawn credit lines plus other loans. Retail loan commitments are excluded from both the numerator and the denominator. *Deposit Base*$_{t-1}$ is the lagged ratio of transaction deposits to total deposits. *Paper-Bill Spread* is the spread on three-month commercial paper rates over Treasuries. High (Low)-*VIX* months are those with the 20% highest (lowest) values of *VIX* in the sample. *T*-statistics are reported in parentheses. Standard errors are clustered by bank. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variable: *Annualized Monthly Average of Bank Squared Returns*							
	High *VIX* periods				Low *VIX* periods			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Commitments$_{t-1}$	0.494**	0.530**	0.534*	0.931**	−0.129	0.007	−0.121	−0.057
	(2.195)	(2.008)	(1.976)	(2.398)	(−0.624)	(0.032)	(−0.509)	(−0.142)
Commitments$_{t-1}$ ×				−2.529				−0.301
Deposit Base$_{t-1}$				(−1.498)				(−0.240)
Deposit Base$_{t-1}$				0.018				0.071
				(0.038)				(0.181)
Log(*VIX*)$_{t-1}$		0.843***	0.846***	0.859***		0.287	0.299	0.304
		(10.166)	(10.226)	(10.431)		(1.413)	(1.547)	(1.615)
Paper-Bill Spread		−0.080	−0.080	−0.075		−0.068	−0.052	−0.050
		(−1.324)	(−1.331)	(−1.255)		(−0.684)	(−0.528)	(−0.531)
Yield on 3-Month T-Bill		0.084***	0.086***	0.092***		−0.043**	−0.034*	−0.034*
		(7.724)	(7.300)	(7.611)		(−2.205)	(−1.791)	(−1.808)
Log of Assets		−0.031	−0.029	−0.030		−0.069***	−0.068***	−0.069**
		(−1.142)	(−1.000)	(−1.120)		(−2.857)	(−2.624)	(−2.471)
(*Cash* + *Securities*)/*Assets*			0.013	0.031			0.388	0.383
			(0.057)	(0.148)			(1.353)	(1.323)
Equity/*Assets*			0.353	0.319			−0.382	−0.389
			(0.828)	(0.776)			(−0.982)	(−1.013)
Constant	−1.226***	−3.948***	−4.041***	−4.093***	−1.573***	−1.020	−1.156*	−1.162*
	(−21.810)	(−7.495)	(−6.981)	(−7.593)	(−25.948)	(−1.643)	(−1.750)	(−1.758)
Observations	1,600	1,600	1,600	1,600	1,590	1,590	1,561	1,561
R^2	0.024	0.143	0.144	0.157	0.002	0.045	0.064	0.064

(a) Table X: Aggregate Risk and Banks' Liquidity Constraints

Table XI

Covenant Violations and Aggregate Risk

This table reports fixed effect regressions of covenant violations on firm characteristics. The dependent variable is *Covenant Violation*, which is a dummy indicating whether there was a covenant violation in that firm-year. All regressions include firm fixed effects. The sample is restricted to firm-years in which the firm has a credit line. Thus, the regressions measure the effect of a change in firm characteristics and *VIX* on the probability of covenant violations. The variables are described in Appendix F. *significant at 10%; **significant at 5%; ***significant at 1%.

	Dependent Variables				
	Covenant Violation				
	(1)	(2)	(3)	(4)	(5)
EBITDA$_t$/*Assets*$_{t-1}$	−0.586***	−0.585***	−0.947	−0.555***	−0.695***
	(−5.143)	(−4.642)	(−1.628)	(−4.299)	(−4.663)
Debt$_t$/*Assets*$_t$	0.402*	0.404	0.401	0.363*	0.341
	(1.791)	(1.642)	(1.627)	(1.670)	(1.583)
Networth$_t$/*Assets*$_t$	−0.175	−0.174	−0.177	−0.154	−0.174
	(−0.829)	(−0.747)	(−0.759)	(−0.727)	(−0.839)
Q	−0.007	−0.006	−0.006	−0.009	−0.010
	(−0.746)	(−0.669)	(−0.630)	(−0.992)	(−1.067)
Size	0.040*	0.040	0.036	0.053**	0.053**
	(1.833)	(1.494)	(1.330)	(2.205)	(2.203)
VIX		0.000	−0.002		
		(0.097)	(−0.352)		
VIX ∗ *EBITDA*$_t$/*Assets*$_{t-1}$			0.014		
			(0.626)		
Beta KMV				−0.012	−0.023
				(−0.955)	(−1.522)
Beta KMV ∗ *EBITDA*$_t$/*Assets*$_{t-1}$					0.103**
					(2.193)
Constant	−0.041	−0.045	0.028	−0.107	−0.070
	(−0.210)	(−0.208)	(0.111)	(−0.504)	(−0.334)
Year fixed effect	Yes	No	No	Yes	Yes
Firm fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	1,425	1,425	1,425	1,186	1,186
R^2	0.103	0.382	0.383	0.096	0.100

revocation (*Revocation Dummy Total*). Columns (4) to (6) replace *Revocation*

(b) Table XI: Covenant Violations and Aggregate Risk

6.10 Table XII: Credit Line Revocations and Aggregate Risk

Read:

- Table XII examines whether covenant violations and aggregate risk predict later reductions in unused credit-line availability or measured revocations.
- Covenant violations predict some subsequent reductions in credit-line availability, consistent with Sufi (2009).
- VIX and beta do not meaningfully predict revocations after controlling for covenant violations and firm changes.

Economic message: Banks price systematic risk ex ante through spreads, fees, maturity, and line availability. They do not appear to revoke credit lines ex post simply because aggregate risk is high.

Table XII

Credit Line Revocations and Aggregate Risk

This table reports regressions of measures of credit line availability on covenant violations, firm characteristics, and *VIX*. The sample is restricted to firm-years in which a credit line was presented in year $t-1$. In Panel A the dependent variable is *Unused LC-to-Assets*, the ratio of unused credit lines to lagged assets. All regressions in Panel A include firm fixed effects. The variables in Panel A are described in Appendix F. In Panel B we consider alternative measures of credit line revocations. In columns (1) and (2) the dependent variable is *Revocation Total*, which is equal to the decrease in the amount of total credit lines in year t relative to year $t-1$, minus the dollar amount of credit lines that matures in year t. In column (3) we use *Revocation Dummy Total*, which is a dummy variable that takes the value of one for firm-years in which *Revocation Total* is positive (that is, there is a reduction in the availability of existing credit lines that have not yet matured). In columns (4) and (5) we use *Revocation Unused*, which is the equivalent of *Revocation Total* but based on changes in unused credit lines. We use *Revocation Dummy Unused* in column (6), which is the equivalent of *Revocation Dummy Total*, but based on changes in unused credit lines. The sample in columns (2), (3), (5), and (6) is restricted to firm-years with no credit line initiations. Firm-level controls are all in differenced form (the change in year t relative to year $t-1$). The data on changes in total and unused credit lines are drawn from Sufi (2009), and described in Appendix F. The data required to compute maturing credit lines and credit line initiations are drawn from LPC-Deal Scan, described in Appendix F. *significant at 10%; **significant at 5%; ***significant at 1%.

Panel A: Credit Line Availability

	Dependent Variable *Unused LC-to-Assets*				
	(1)	(2)	(3)	(4)	(5)
Covenant Violation$_{t-1}$	−0.041***	−0.041***	−0.081	−0.040***	−0.043***
	(−3.411)	(−3.041)	(−0.719)	(−3.768)	(−3.221)
EBITDA$_{t-1}$/*Assets*$_{t-2}$	0.095**	0.091*	0.090*	0.062	0.062
	(2.135)	(1.784)	(1.775)	(1.355)	(1.356)
Debt$_{t-1}$/*Assets*$_{t-1}$	0.048	0.040	0.040	0.064	0.063
	(0.365)	(0.266)	(0.266)	(0.406)	(0.396)
Networth$_{t-1}$/*Assets*$_{t-1}$	0.048	0.041	0.042	−0.012	−0.013
	(0.503)	(0.392)	(0.393)	(−0.156)	(−0.171)
Q_{t-1}	0.007	0.007	0.007	0.007	0.007
	(1.182)	(1.001)	(0.997)	(1.027)	(1.029)
Size$_{t-1}$	−0.120***	−0.117***	−0.117***	−0.089***	−0.089***
	(−4.143)	(−3.501)	(−3.511)	(−3.465)	(−3.464)
VIX$_{t-1}$		−0.001	−0.001		
		(−0.495)	(−0.552)		
VIX$_{t-1}$ ∗ *Covenant Violation*$_{t-1}$			0.002		
			(0.355)		
Beta KMV$_{t-1}$				0.001	0.001
				(0.229)	(0.138)
Beta KMV$_{t-1}$ ∗ *Covenant Violation*$_{t-1}$					0.004
					(0.529)
Constant	0.754***	0.760***	0.762***	0.617***	0.618***
	(5.435)	(4.907)	(4.897)	(4.459)	(4.452)
Year fixed effect	Yes	No	No	Yes	Yes
Firm fixed effect	Yes	Yes	Yes	Yes	Yes
Observations	1,206	1,206	1,206	1,010	1,010
R^2	0.170	0.607	0.607	0.131	0.131

(Continued)

(a) Table XII: Credit Line Revocations and Aggregate Risk, Panel A

Table XII—*Continued*

Panel B: Credit Line Revocation

	Dependent Variables					
	Revocation Total		*Revocation Dummy Total*	*Revocation Unused*		*Revocation Dummy Unused*
	(1)	(2)	(3)	(4)	(5)	(6)
Covenant Violation$_{t-1}$	0.044**	0.039*	0.178***	0.009	0.013	0.088
	(2.104)	(1.815)	(2.780)	(0.502)	(0.731)	(1.478)
Vix$_{t-1}$	0.001	0.001	0.007	0.001	−0.000	−0.001
	(0.672)	(0.532)	(1.413)	(0.468)	(−0.093)	(−0.163)
Δ*EBITDA*$_{t-1}$/*Assets*$_{t-1}$	0.093	0.050	0.072	−0.022	−0.006	−0.155
	(1.227)	(0.682)	(0.373)	(−0.316)	(−0.078)	(−0.775)
Δ*Debt*$_{t-1}$/*Assets*$_{t-1}$	0.340***	0.250**	0.545	0.050	−0.033	0.060
	(3.215)	(2.201)	(1.639)	(0.488)	(−0.299)	(0.142)
Δ*Networth*$_{t-1}$/*Assets*$_{t-1}$	0.084	0.069	0.058	−0.028	−0.047	−0.085
	(0.870)	(0.632)	(0.220)	(−0.264)	(−0.398)	(−0.259)
ΔQ_{t-1}	−0.015**	−0.010	−0.041**	−0.004	−0.007	−0.003
	(−2.220)	(−1.566)	(−2.070)	(−0.717)	(−1.084)	(−0.138)
Δ*Size*$_{t-1}$	−0.086***	−0.057**	−0.103	−0.001	−0.009	−0.077
	(−3.816)	(−2.336)	(−1.447)	(−0.033)	(−0.440)	(−1.337)
Constant	−0.092*	−0.078	0.123	−0.072*	−0.039	0.396***
	(−1.871)	(−1.363)	(0.955)	(−1.882)	(−0.870)	(3.014)
Observations	877	700	700	877	700	700
R^2	0.038	0.025	0.042	0.005	0.003	0.010

(b) Table XII: Credit Line Revocations and Aggregate Risk, Panel B

7 Short summary

- The paper develops a theory in which credit lines are bank-provided liquidity insurance and cash is self-insurance.
- Banks can diversify idiosyncratic liquidity shocks but cannot fully diversify aggregate drawdown risk.
- Therefore, high-beta firms should hold more cash and rely less on credit lines.
- Empirically, beta is negatively related to LC-to-Cash across multiple beta measures, samples, and specifications.
- The beta effect is strongest among financially constrained firms and high-beta firms, as predicted by the model.
- High-beta firms face higher costs on credit lines, especially higher fees on undrawn balances.
- When VIX or Bank VIX rises, credit line initiations fall, credit line maturities shorten, spreads rise, and cash holdings increase.

- Bank-level tests show that unused credit-line commitments increase bank risk in high-VIX periods.
- Covenant-violation tests do not support the interpretation that the main findings are driven by systematic differences in line revocation risk.

8 Conclusion

8.1 Core conclusion

Aggregate risk is central to firms' liquidity management. Firms exposed to systematic liquidity shocks rely less on bank credit lines and more on cash. This happens because bank lines of credit are efficient for pooling idiosyncratic risk but costly when drawdowns are correlated across firms.

8.2 Economic intuition

A bank can write liquidity insurance cheaply if only a few firms draw at once. But if many firms need liquidity in the same state, the bank must either hold costly liquidity itself or charge more for commitments. High-beta firms internalize this cost through worse credit-line terms and therefore use cash.

8.3 Incremental contribution revisited

The paper connects corporate liquidity policy to aggregate risk and intermediary balance-sheet capacity. It shows that the cash-credit-line choice is not just a firm-level precautionary savings decision; it is also a question of how aggregate risk is allocated between firms and banks.

8.4 Implications for corporate finance

- Cash is particularly valuable for firms whose financing needs arise in bad aggregate states.
- Credit lines are most useful for firms whose liquidity shocks are idiosyncratic and insurable by banks.
- Firms should expect credit-line terms to tighten in high-volatility periods, even if their own fundamentals have not deteriorated sharply.

8.5 Implications for banks and financial stability

Undrawn corporate credit lines create contingent liquidity risk for banks. This risk is muted in normal times but becomes important when aggregate volatility is high and many firms may draw simultaneously. Monitoring the stock of undrawn commitments is therefore important for assessing banking-sector fragility.

8.6 Limitations

- The empirical design is observational, so results rely on controls, instruments, robustness checks, and the coherence of multiple predictions rather than random assignment.
- Beta is an imperfect proxy for liquidity-shock correlation, although the paper uses many alternative beta measures.
- The LPC-DealScan sample is tilted toward large syndicated credit-line deals.
- The model abstracts from richer bank-capital dynamics and the possibility of government support in crises.

8.7 Takeaway

The paper's main lesson is that "cash is king" precisely when aggregate risk makes bank liquidity insurance fragile. Credit lines are valuable commitments, but commitments are only as strong as the intermediary sector's ability to honor them when many firms need cash at the same time.